Settings in a Tab: Click Heatmap Report

Click data analysis of the Lyssna prototype test · October 2026

Introduction

In this task I looked at click data from an unmoderated Lyssna test of a new settings prototype. 76 developers had to open the settings, turn on “Use contrast scrollbars” and go back to the code. Compared to the current design, the prototype changes two things:

1. Settings open as a tab in the editor instead of a separate window.
2. When you change something, a popup asks you to save or revert. There are no OK / Cancel / Apply buttons anymore.

Short version: almost everyone managed to turn the setting on and save it, but after that a lot of people got stuck. 60 of 76 (79%) saved the change, while Lyssna counted only 19 (25%) as successful.

Data and what I did with it

The export has 76 respondents, each with one task, and 851 clicks in total on these screens:

Screen ID	Screen	What the user should do
1:12610	Main Window	Click the gear icon (top right)
1:12676	Context menu	Choose “Settings…”
1:12541	Default	Tick “Use contrast scrollbars”
1:12401	Changed	Click “Save Changes” in the popup
1:12472	Saved	Go back to the code (FunctionUtils.java tab)
1:12624	Code	Goal screen, no clicks here

I wrote the analysis in Python (NumPy, SciPy, Matplotlib, Pillow). At first it was one script, but it got messy, so I split it into a few files:

- study.py has a Study class that loads the JSON, collects clicks per screen and calculates the stats (funnel, where people gave up, misclicks, first clicks, median times)

- heatmaps.py draws the heatmaps
- saliency.py compares first clicks with a simple saliency model (more on that later)
- screens.py has the screen ids and finds the screenshot for each screen
- main.py runs everything

For the heatmaps I count clicks per pixel, blur them with a Gaussian filter (sigma = 1.5% of the screen width) and draw that on top of the screenshot. Every screen gets two heatmaps, one with all clicks and one with just the first click of each person. Luckily the click coordinates were already in the same pixel space as the screenshots, so no rescaling was needed. I also counted, for each screen, how many people clicked the right thing on their first try.

One more screen, 1:12650, showed up in the data. Only 2 people reached it and there is just one click on it, so I didn't put its heatmap in the report. Looking at the paths, it's probably the "unsaved changes" state you get when you try to leave settings without saving (both people came to it from the "Changed" screen).

Overall results

69% of all clicks (587 of 851) were misclicks. The funnel looks fine until people save. The real drop comes after that, when they need to get out of the settings:

Screen	People who reached it
Main Window	76 (100%)
Context menu	68 (89%)
Default (settings open)	68 (89%)
Changed (checkbox ticked)	64 (84%)
Saved	60 (79%)
Code	19 (25%)

Part of why the Lyssna number is so low is how success was defined. It only counts if you reach the "Code" screen by clicking the code tab. 24 people saved and then closed the Settings tab instead. In an IDE that's a completely normal way to go back to your code, but in the prototype closing the tab takes you back to the first screen, so Lyssna marked them as "gave up". If I count them, 40 of 76 people (53%) actually did the whole task.

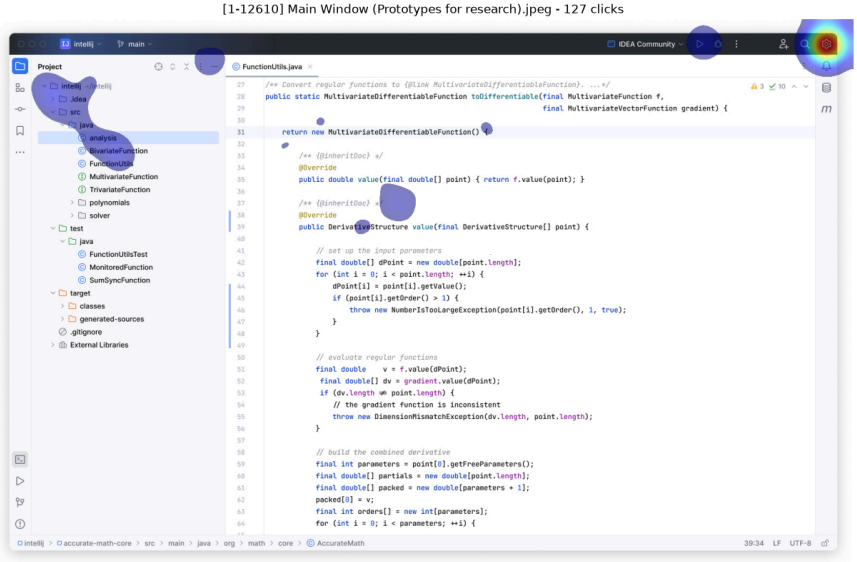
That leaves 20 people who saved but never got back to the code. 17 of them just ended the task on the Settings tab, and 3 went back and forth for a while and then gave up. Of the 16 who never saved, 8 never even opened the gear menu and 5 got into settings but didn't save. The last 3 are actually counted as "completed", because they switched to the code tab without saving. Median time was 77 s for people who completed and 58 s for people who gave up.

Findings per screen

On the heatmaps, warmer color means more clicks in that spot. The title of each image has the number of clicks. Each heatmap is scaled to its own maximum, so colors can't be compared between screens.

1. Main window

The gear is the hottest spot, but only 32 of the 59 people with recorded clicks hit it first, and 8 of 76 never opened the menu at all. The rest of the clicks are mostly in the code editor and the project tree. A few people also clicked Run and Debug, which are right next to the gear.

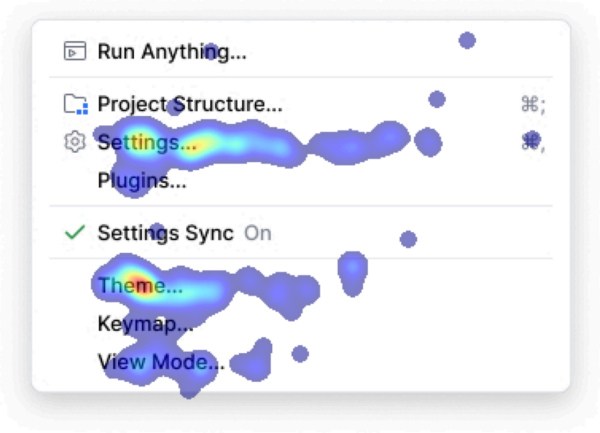


Main window heatmap

2. Gear menu

43 of 68 (63%) picked "Settings…" first. The other big hot spot is "Theme…", and some people also tried "View Mode…". My guess is that it's because of the wording: the task said "change the appearance", and "Theme" sounds a lot closer to appearance than "Settings".

[1-12676] Comtext menu (Prototypes for research).jpeg - 161 clicks

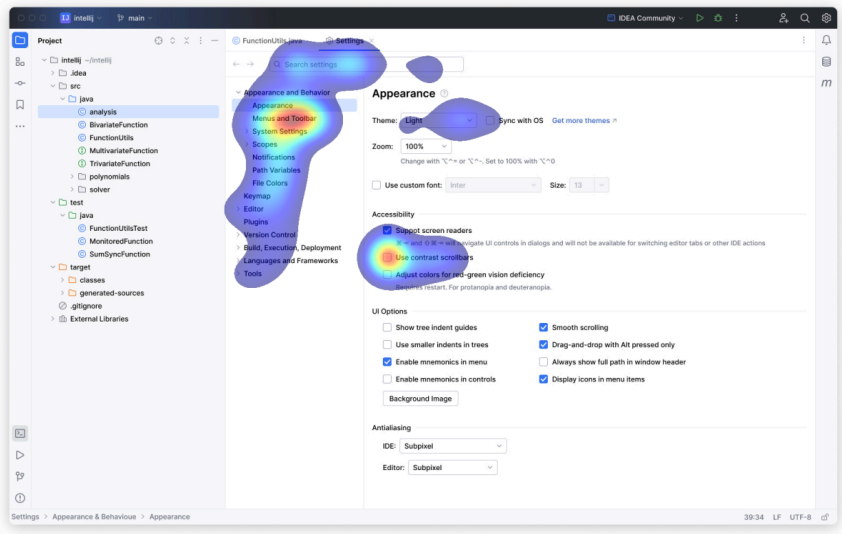


Gear menu heatmap

3. Settings page

This screen had the most clicks of all (323). Settings already open on the Appearance page and the checkbox is right there, but only 21 of 65 clicked it first. Most of the other clicks went to the category tree on the left and to the search bar. To be honest, that's exactly what I'd do in a real IDE too. In the prototype neither of them works, though, so all of it counts as misclicks.

[1-12541] Default (Prototypes for research).jpeg - 323 clicks

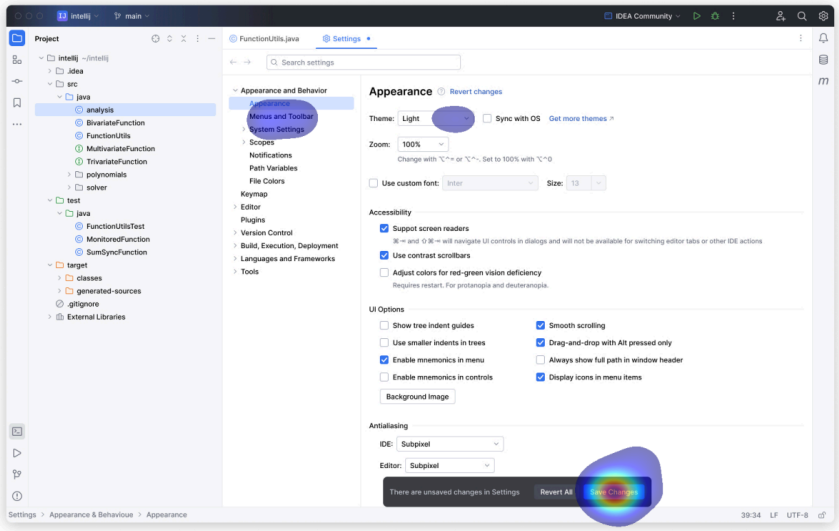


Settings page heatmap

4. After ticking the checkbox

This step went well. 57 of 64 (89%) clicked "Save Changes" first, so the popup is easy to spot. Apart from the Save button there's almost nothing around the popup on the heatmap, just a few clicks on the settings tree and the theme dropdown.

[1-12401] Changed (Prototypes for research).jpeg - 100 clicks

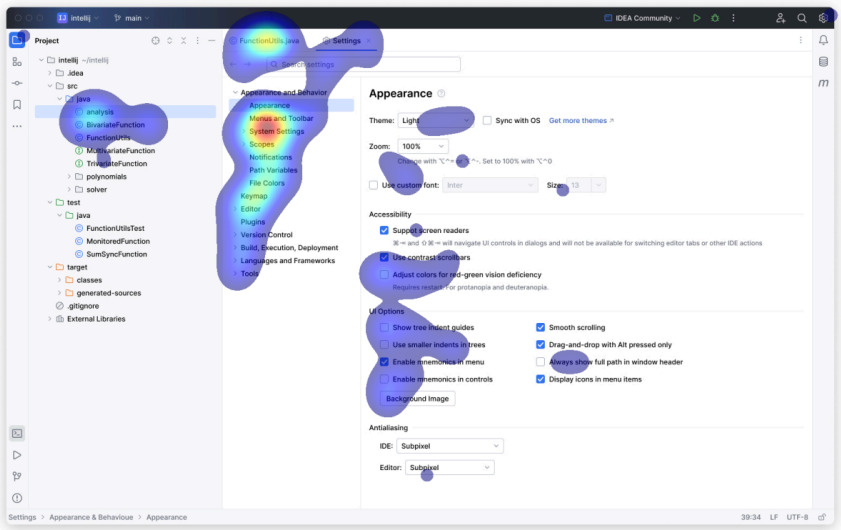


Changed state heatmap

5. After saving

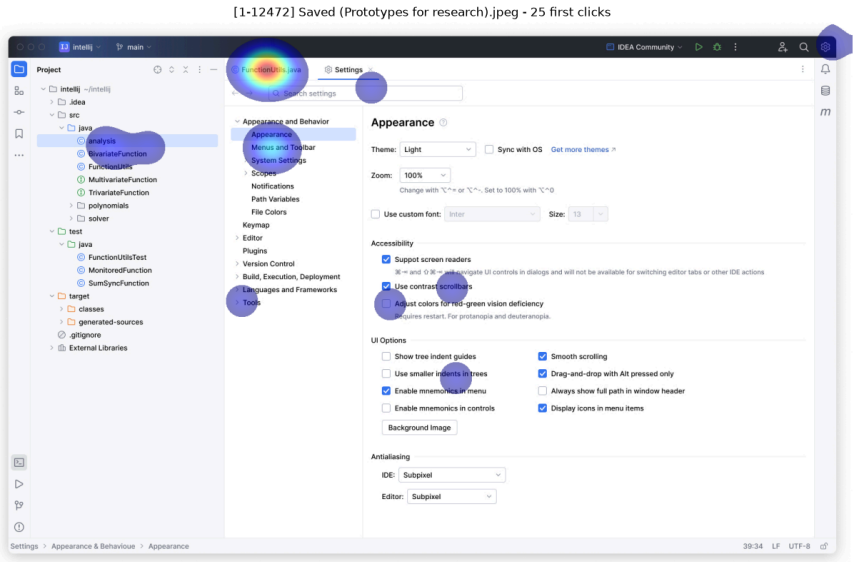
This is where people got lost. The way back is the FunctionUtils.java tab, and it is the hottest spot, but only 12 of the 25 people who clicked here hit a hotspot first. Others clicked around in the category tree, the project tree and the settings. Most people on this screen didn't click at all, they either closed the Settings tab or just ended the task.

[1-12472] Saved (Prototypes for research).jpeg - 139 clicks



Saved state heatmap

The first-click heatmap makes this clearer. Most first clicks went to the FunctionUtils.java tab, so the people who actually looked for a way back mostly found it.



Saved state, first clicks only

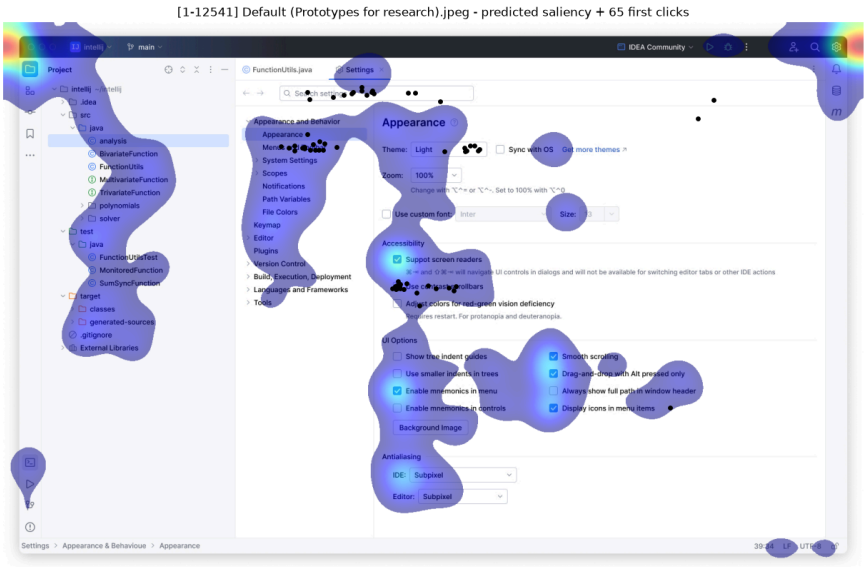
Do first clicks go to what stands out?

I was also curious whether people click first on whatever stands out visually, or whether they go for what they're looking for. To check this I made a saliency map for every screenshot using the spectral residual method (Hou & Zhang, 2007), which is a simple model that marks the parts of an image that "pop out". Then I counted how many first clicks fall into the top 20% of that map. To see if saliency adds anything at all, I did the same with a plain edge map (basically where there's a lot of contrast). With random clicks you'd expect around 20%.

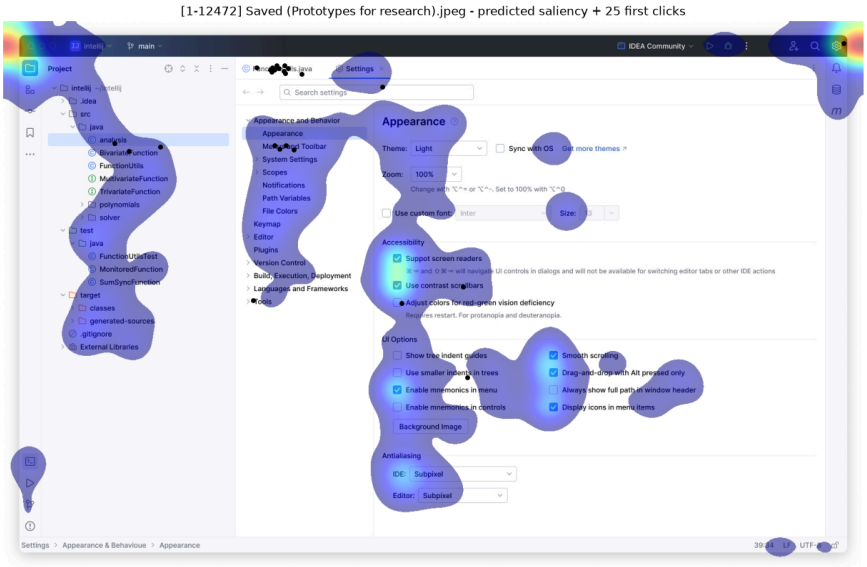
Screen	First clicks	In top 20% (saliency)	In top 20% (edges only)
Main Window	59	64%	69%
Context menu	68	43%	43%
Default	65	72%	54%
Changed	64	92%	94%
Saved	25	44%	92%

On most screens first clicks land in the "visible" parts much more often than chance would give, but saliency isn't really better than plain edges. Default is the only screen where it helps (72% vs 54%), because the checkbox area stands out. On the Saved screen it's the other way around: first clicks went to the tabs at the top, which have lots of edges, but the saliency model doesn't pick them up. In the menu both maps highlight all items about the same, so neither can explain why people chose "Theme…" over "Settings…". I skipped 1:12650 here because it has only one first click.

So from what I can see, people don't just click on what stands out. They mostly go for what matches the task. Saliency can't explain mistakes like "Theme…" or searching through the category tree.



Saliency and first clicks on the Default screen



Saliency and first clicks on the Saved screen

What this means for the two changes

The save / revert popup (change 2) seems to work. 89% clicked "Save Changes" first and there are almost no misclicks around it, so people notice and understand it even without OK / Cancel / Apply.

Settings in a tab (change 1) is where the problem is: there's no clear way out. In the old dialog, OK saved and closed everything in one click. Now you save and you're still on the Settings tab, and closing it is a separate step that nothing points you to. I saw two main reactions:

- 17 people stopped right after saving, probably thinking they were done.
- 24 people closed the tab to get back to the code, which is what you'd normally do in an editor.

Recommendations

1. Give people a clear next step after "Save Changes", for example a "Saved" message with a "Close settings" button. Another option is to close the tab automatically if it was opened just to change one setting.
2. Show that the setting is on now (for example, briefly highlight the scrollbar), so people know it worked and the task is done.
3. Make "Settings…" easier to tell apart from "Theme…" and "View Mode…" in the menu, or let "Theme…" open the Appearance page too.
4. In the next test, count "setting saved and back in the code" as success, no matter if people got back by clicking the code tab or by closing Settings. It would also help to make the category tree and search clickable, since that's how developers normally look for settings.

Limitations

- It's a prototype, not the real product. Only hotspots do anything, so search, the category tree and other controls don't work. Many of the misclicks on the settings screens are probably just normal behavior.
- For some people a screen change has no recorded click (for example, the menu opened but there's no click on the gear), so the first-click numbers only use the clicks that were recorded.
- Clicks are stored per screen, not per visit. If someone visited a screen twice, the clicks from both visits are mixed, so I can't always tell exactly how they got back to the main window.
- The success definition is quite narrow. 3 of the 19 "completed" people never actually saved the setting.
- The saliency model is very simple and was made for natural images, not UI screenshots, so those numbers are only a rough check.
- There was no think-aloud and no follow-up questions. Why people stopped after saving is my guess based on the paths, and it would be worth checking in a few moderated sessions.